

Rutvik Shah

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SUMMARY

ML Engineer and Data Analyst with **3.5 years of experience** driving data-driven decisions through building and deploying predictive models, optimizing workflows, and creating actionable insights. Proven track record of improving accuracy, reducing processing time, and enabling faster, more informed decision-making across diverse business and analytical projects.

TECHNICAL SKILLS

Programming & ML: Python (Pandas, NumPy, SciPy, TensorFlow, PyTorch, scikit-learn, Spark ML, Caffe2), Java, JavaScript, Flask, AI/ML Programming, Data Structures & Algorithms, Object-Oriented Programming

Data Analytics & Visualization: Power BI, Tableau, SQL Server, MySQL, PostgreSQL, Microsoft Excel, Exploratory Data Analysis (EDA), A/B Testing, Data Cleaning, Data Transformation, Statistical Analysis

Machine Learning & AI Techniques: Generative AI (Claude 3.5, Llama-7B-instruct), Sentence Transformers, AdaBoost, Random Forest, Gradient Boosting, PCA, UMAP, K-means, Hierarchical Clustering, Embedding Methods, Cosine Similarity, Model Evaluation & Hyperparameter Tuning

Big Data & Tools: Apache Spark, PySpark, Jupyter Notebook, Git/GitHub, Pickle for caching & serialization, RESTful API integration, JSON, CSV, Document Parsing (PDF/DOCX)

Other / Cloud & Productivity: HTML, CSS (Tailwind), Jinja2 Templates, Session Management, Version Control (Git), Agile Methodologies, Problem-Solving & Debugging

PROFESSIONAL EXPERIENCE

Intuit, USA Jan 2025 – Current

Associate ML Engineer

- Built and deployed ML models using Python, TensorFlow, PyTorch, and Spark ML, improving prediction accuracy by 18%, enabling faster, data-driven decisions across multiple business teams.
- Streamlined data pipelines for large datasets using Pandas, NumPy, and PySpark, reducing processing time by 30% and increasing workflow efficiency for analytics and engineering teams.
- Integrated ML solutions with RESTful APIs and implemented Pickle-based caching, cutting production delays by 25% and accelerating deployment of predictive models in real-time environments.
- Created interactive dashboards in Power BI and Tableau, providing actionable insights that improved cross-team visibility, enabling 15% faster decisions and better operational monitoring.

Cognizant, India Aug 2022 – Jul 2023

Data Analyst & ML Engineer

- Designed and implemented ML models using Python, scikit-learn, and Spark ML, improving project efficiency by 22% and enhancing accuracy for client-facing predictive analytics solutions.
- Conducted exploratory data analysis, statistical testing, and A/B experiments to identify patterns, increasing insight reliability by 18% and guiding strategic business decisions effectively.
- Automated data cleaning and transformation pipelines for CSV, JSON, and PDF files, saving over 10 hours weekly, improving team productivity and project turnaround time significantly.
- Developed dashboards and SQL reports that enabled stakeholders to make faster, data-backed decisions, improving reporting speed and positively impacting overall project delivery timelines.

Deloitte, India Aug 2021 – Jul 2022

Data Analyst

- Cleaned, analyzed, and visualized client datasets using Python, SQL, and Excel, reducing reporting errors by 15% and improving overall data quality and accuracy.
- Applied K-means, hierarchical clustering, and cosine similarity to segment data, improving targeting strategies by 12% and enabling more effective client campaigns and insights.
- Collaborated with stakeholders to translate analytical findings into actionable strategies, improving decision-making speed and enhancing operational efficiency across multiple client projects.
- Developed dashboards and visual reports that presented complex data simply, enabling teams to understand insights quickly and make data-informed decisions efficiently.

Deloitte, India Feb 2021 – Jul 2021

Intern Research Analyst

- Assisted in research and preprocessing of datasets using Python and Jupyter Notebook, improving AI/ML project readiness by 20% and ensuring high-quality input data.
- Explored, cleaned, and organized raw data for AI/ML models, reducing manual work by 15 hours per month and accelerating data preparation processes.
- Supported senior analysts by integrating datasets into Tableau dashboards, enabling clear visualization and actionable insights for internal teams and client presentations.
- Documented research findings and analytical results systematically, improving team knowledge sharing and ensuring efficient tracking of data-driven insights throughout project lifecycles.

EDUCATION

Master of Computer Science in concentration of Artificial Intelligence Hofstra University, New York, USA	Aug 2023 – May 2025
Bachelor of Engineering in Computer Engineering Gujarat Technological University, Ahmedabad, India	Jun 2018 – May 2022

PROJECTS

Body Performance Analysis: Unsupervised Learning

- Applied Random Forest and AdaBoost models, PCA and UMAP dimensionality reduction, K-means and Hierarchical Clustering, achieving 85.94% accuracy and actionable insights for predictive performance analysis.
- Delivered comprehensive data structure understanding, optimized clustering accuracy, and improved pattern recognition, enabling informed decision-making and enhancing predictive model reliability in body performance analytics.

RAG-Augmented-NLP-for-Intelligent-Property-Recommendations

- Built embedding-based recommendation engine using Sentence Transformer and cosine similarity, integrating Llama-7B-instruct for tailored suggestions, improving query relevance by 30% and accelerating response times.
- Implemented cached embeddings with Pickle, optimizing retrieval efficiency, enhancing user experience, and aligning property recommendations accurately with user searches for intelligent accommodation suggestions.

NeuraLearn Tutor: AI-Powered Learning Assistant

- Developed AI-powered tutoring system using Claude 3.5 LLM, delivering personalized, real-time responses and lesson plans while managing multi-turn student sessions efficiently with Flask and Pickle.
- Engineered responsive UI with HTML, Tailwind, and Jinja2 templates, implementing session analytics achieving 1.8s GenAI latency and 4.8/5 user satisfaction for adaptive learning experiences.

PUBLICATIONS

Machine Learning in Medicine: Diagnosis of Skin Cancer using Support Vector Machine (SVM) Classifier	May 2022
- Conducted a comprehensive review of 300+ studies on melanoma detection using image processing techniques. - Investigated advanced SVM-based classification methods, improving diagnostic accuracy by 30% and supporting AI-driven early, non-invasive skin cancer detection.	